Pre-Trained Model Datasets Tables

Table 1: Compound dataset formed cutoff with X =1000 nM

Target Family	Total Compound	Active compound	Inactive compound
GPCR Subfamily A3	1632	1216	416
Nuclear Receptor Subfamily 3	7049	5681	1368
GPCR Subfamily A17	22533	19347	3186
Total	31214	26244	4970

Table 2: Compound dataset formed cutoff with X =500 nM

Target Family	Total Compound	Active compound	Inactive compound
GPCR Subfamily A3	1632	1113	519
Nuclear Receptor Subfamily 3	6928	5137	1791
GPCR Subfamily A17	21301	16837	4464
Total	29861	23087	6774

Table 3: Compound dataset formed cutoff with X =300 nM

Target Family	Total Compound	Active compound	Inactive compound
GPCR Subfamily A3	1630	1021	609
Nuclear Receptor Subfamily 3	6855	4719	2136
GPCR Subfamily A17	20744	15200	5544
Total	29229	20940	8289

Table 4: Compound dataset formed cutoff with X = 100 nM

Target Family	Total Compound	Active compound	Inactive compound
GPCR Subfamily A3	1630	861	769
Nuclear Receptor Subfamily 3	6830	3854	2976
GPCR Subfamily A17	19747	11627	8120
Total	28207	16342	11865

Multi-Task Model AUC Score Results

Table 5: Pre-Trained model/Multitask Model resulting AUC task scores at k = 50, h = 5, n = 1024

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
50	**5**	**1024**	**1000**	**GPCR Subfamily A3**	**0.940**	**0.858**
				Nuclear Receptor Subfamily 3	**0.853**	
				GPCR Subfamily A17	**0.780**	
50	5	1024	500	GPCR Subfamily A3	0.899	0.838
				Nuclear Receptor Subfamily 3	0.859	
				GPCR Subfamily A17	0.756	
50	5	1024	300	GPCR Subfamily A3	0.890	0.825
				Nuclear Receptor Subfamily 3	0.840	
				GPCR Subfamily A17	0.745	
50	5	1024	100	GPCR Subfamily A3	0.911	0.847
				Nuclear Receptor Subfamily 3	0.859	
				GPCR Subfamily A17	0.771	

*k = embedding size, h = window size of filter, n = size of feature map

Table 6: Pre-Trained model/Multitask Model resulting AUC task scores at k = 100, h = 5, n = 1024

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
100	5	1024	1000	GPCR Subfamily A3	0.869	0.831
				Nuclear Receptor Subfamily 3	0.857	
				GPCR Subfamily A17	0.766	
100	5	1024	500	GPCR Subfamily A3	0.892	0.849
				Nuclear Receptor Subfamily 3	0.864	
				GPCR Subfamily A17	0.790	
100	5	1024	300	GPCR Subfamily A3	0.863	0.815
				Nuclear Receptor Subfamily 3	0.809	
				GPCR Subfamily A17	0.774	
100	5	1024	100	GPCR Subfamily A3	0.862	0.822
				Nuclear Receptor Subfamily 3	0.847	
				GPCR Subfamily A17	0.758	

*k = embedding size, h = window size of filter, n = size of feature map

Table 7: Pre-Trained model/Multitask Model resulting AUC task scores at k = 150, h = 5, n = 1024

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
150	5	1024	1000	GPCR Subfamily A3	0.958	0.857
				Nuclear Receptor Subfamily 3	0.847	
				GPCR Subfamily A17	0.766	
150	5	1024	500	GPCR Subfamily A3	0.912	0.840
				Nuclear Receptor Subfamily 3	0.832	
				GPCR Subfamily A17	0.776	
150	5	1024	300	GPCR Subfamily A3	0.859	0.819
				Nuclear Receptor Subfamily 3	0.803	
				GPCR Subfamily A17	0.794	
150	5	1024	100	GPCR Subfamily A3	0.870	0.830
				Nuclear Receptor Subfamily 3	0.850	
				GPCR Subfamily A17	0.771	

*k = embedding size, h = window size of filter, n = size of feature map

Table 8: Pre-Trained model/Multitask Model resulting AUC task scores at k = 50, h = 3, n = 1024

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
50	3	1024	1000	GPCR Subfamily A3	0.886	0.834
				Nuclear Receptor Subfamily 3	0.863	
				GPCR Subfamily A17	0.754	
50	3	1024	500	GPCR Subfamily A3	0.929	0.858
				Nuclear Receptor Subfamily 3	0.878	
				GPCR Subfamily A17	0.766	
50	3	1024	300	GPCR Subfamily A3	0.895	0.832
				Nuclear Receptor Subfamily 3	0.825	
				GPCR Subfamily A17	0.777	
50	3	1024	100	GPCR Subfamily A3	0.927	0.837
				Nuclear Receptor Subfamily 3	0.824	
				GPCR Subfamily A17	0.761	

*k = embedding size, h = window size of filter, n = size of feature map

Table 9: Pre-Trained model/Multitask Model resulting AUC task scores at k = 50, h = 4, n = 1024

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
50	4	1024	1000	GPCR Subfamily A3	0.937	0.851
				Nuclear Receptor Subfamily 3	0.848	
				GPCR Subfamily A17	0.769	
50	4	1024	500	GPCR Subfamily A3	0.903	0.839
				Nuclear Receptor Subfamily 3	0.855	
				GPCR Subfamily A17	0.758	
50	4	1024	300	GPCR Subfamily A3	0.908	0.851
				Nuclear Receptor Subfamily 3	0.861	
				GPCR Subfamily A17	0.784	
50	4	1024	100	GPCR Subfamily A3	0.866	0.825
				Nuclear Receptor Subfamily 3	0.836	
				GPCR Subfamily A17	0.772	

*k = embedding size, h = window size of filter, n = size of feature map

Table 10: Pre-Trained model/Multitask Model resulting AUC task scores at k = 50, h = 5, n = 256

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
50	5	256	1000	GPCR Subfamily A3	0.857	0.799
				Nuclear Receptor Subfamily 3	0.799	
				GPCR Subfamily A17	0.741	
50	5	256	500	GPCR Subfamily A3	0.915	0.842
				Nuclear Receptor Subfamily 3	0.844	
				GPCR Subfamily A17	0.766	
50	5	256	300	GPCR Subfamily A3	0.862	0.820
				Nuclear Receptor Subfamily 3	0.843	
				GPCR Subfamily A17	0.755	
50	5	256	100	GPCR Subfamily A3	0.926	0.827
				Nuclear Receptor Subfamily 3	0.825	
				GPCR Subfamily A17	0.729	

*k = embedding size, h = window size of filter, n = size of feature map

Table 11: Pre-Trained model/Multitask Model resulting AUC task scores at k = 50, h = 5, n = 512

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
50	5	512	1000	GPCR Subfamily A3	0.923	0.846
				Nuclear Receptor Subfamily 3	0.849	
				GPCR Subfamily A17	0.766	
50	5	512	500	GPCR Subfamily A3	0.939	0.850
				Nuclear Receptor Subfamily 3	0.842	
				GPCR Subfamily A17	0.769	
50	5	512	300	GPCR Subfamily A3	0.879	0.822
				Nuclear Receptor Subfamily 3	0.826	
				GPCR Subfamily A17	0.761	
50	5	512	100	GPCR Subfamily A3	0.900	0.833
				Nuclear Receptor Subfamily 3	0.843	
				GPCR Subfamily A17	0.756	

*k = embedding size, h = window size of filter, n = size of feature map

Single-Task Model Results

Table 11: Single Task resulting AUC task scores at k = 50, h = 5, n = 1024

Hyperparameters			N (nM)	Target Protein Family	AUC Scores	Average AUC Score
k	h	n				
50	5	**1024**	1000	GPCR Subfamily A3	0.923	0.849
				Nuclear Receptor Subfamily 3	0.824	
				GPCR Subfamily A17	0.802	
50	5	1024	500	GPCR Subfamily A3	0.878	0.819
				Nuclear Receptor Subfamily 3	0.829	
				GPCR Subfamily A17	0.751	
50	5	1024	300	GPCR Subfamily A3	0.846	0.815
				Nuclear Receptor Subfamily 3	0.832	
				GPCR Subfamily A17	0.767	
50	5	1024	100	GPCR Subfamily A3	0.884	0.818
				Nuclear Receptor Subfamily 3	0.803	
				GPCR Subfamily A17	0.767	

*****k = embedding size, h = window size of filter, n = size of feature map

Logarithm Loss

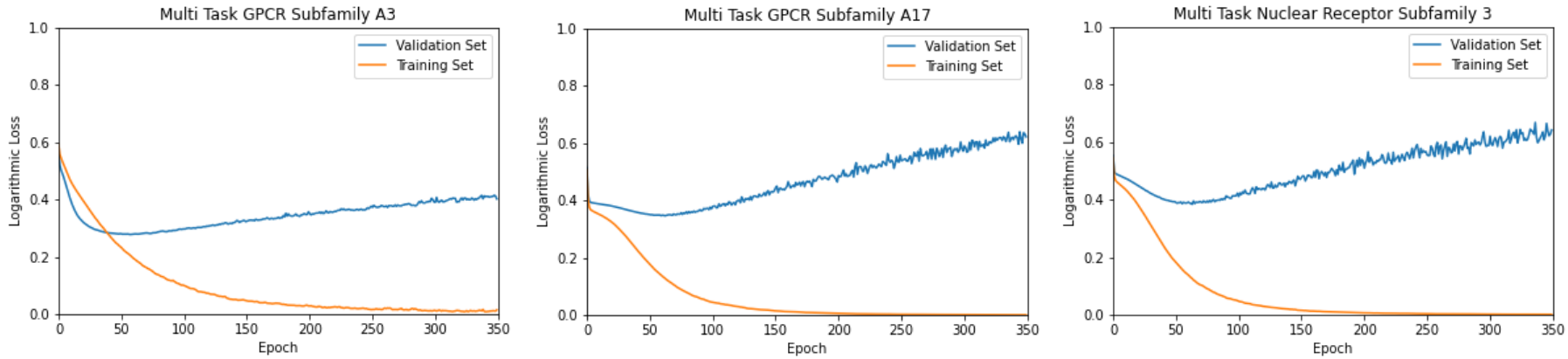

Figure 1: Relation between **Logarithmic Loss** and **Epoch** for GPCR Subfamily A3 (left), Nuclear Receptor Subfamily 3 (middle) and GPCR Subfamily A17 (right) protein target family tasks with k = 50, h = 5, n = 1024, N = 500 nM for Pre-Trained/Multi-task model

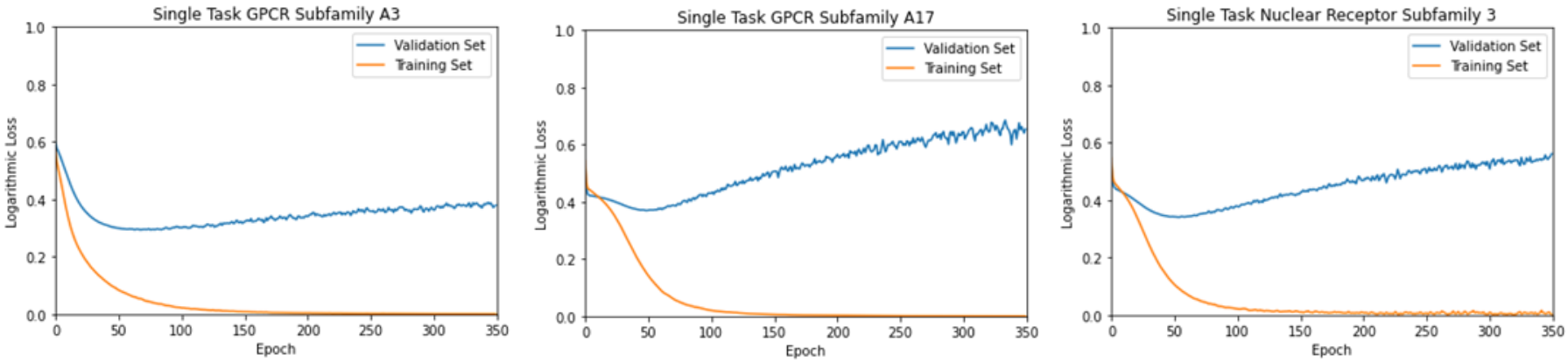

Figure 2: Relation between **Logarithmic Loss** and **Epoch** for GPCR Subfamily A3 (left), Nuclear Receptor Subfamily 3 (middle) and GPCR Subfamily A17 (right) protein target family tasks with k = 50, h = 5, n = 1024, N = 500 nM for Single task model